

Deterministic and Non-Deterministic Functions in SQL Server

In this article, I am going to discuss **Deterministic and Non-Deterministic Functions in SQL Server** with Examples. Please read our previous article where we discussed [Encryption and Schema Binding Option in SQL Server](#). At the end of this article, you will understand what are Deterministic and Non-Deterministic Functions in SQL Server.

Example: Deterministic and Non-Deterministic Functions in SQL Server

We are going to use the following Employee table to understand Deterministic and Non-Deterministic Functions in SQL Server.

ID	Name	Gender	DOB	DeptID
1	Pranaya	Male	1996-02-29 10:53:27.060	1
2	Priyanka	Female	1995-05-25 10:53:27.060	2
3	Anurag	Male	1995-04-19 10:53:27.060	2
4	Preety	Female	1996-03-17 10:53:27.060	3
5	Sambit	Male	1997-01-15 10:53:27.060	1
6	Hina	Female	1995-07-12 10:53:27.060	2

Please use the below script to create and populate the Employee table with sample data.

```
-- Create Employee Table
CREATE TABLE Employee
(
    ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Gender VARCHAR(50),
    DOB DATETIME,
    DeptID INT
)
GO
-- Populate the Employee Table with test data
INSERT INTO Employee VALUES(1, 'Pranaya', 'Male', '1996-02-29 10:53:27.060', 1)
INSERT INTO Employee VALUES(2, 'Priyanka', 'Female', '1995-05-25 10:53:27.060', 2)
INSERT INTO Employee VALUES(3, 'Anurag', 'Male', '1995-04-19 10:53:27.060', 2)
INSERT INTO Employee VALUES(4, 'Preety', 'Female', '1996-03-17 10:53:27.060', 3)
INSERT INTO Employee VALUES(5, 'Sambit', 'Male', '1997-01-15 10:53:27.060', 1)
INSERT INTO Employee VALUES(6, 'Hina', 'Female', '1995-07-12 10:53:27.060', 2)
GO
```

Deterministic Functions in SQL server

The Deterministic functions in SQL Server always return the same result every time they are called with a specific set of input values and give the same state of the database. For examples: **SQUARE()**, **Power()**, **Sum()**, **AVG()**, and **Count()**. All aggregate functions are deterministic functions. Let us see some examples.

Example1: `SELECT COUNT(*) FROM Employee`

Every time we execute the above statement it will give 6

Example2: `SELECT SQUARE(3)`

Every time we execute the above statement it will give 9

Non-Deterministic Functions in SQL Server

The Non-deterministic functions in SQL Server may return different results each time they are called with a specific set of values even if the database state that they access remains the same. For examples: `GetDate()` and `Current_Timestamp`

Example1: `SELECT GetDate()`

Every time we execute the above SQL Query, it will give a different result

Example2: `SELECT Current_timestamp`

Every time the results will vary, it will return the date and time of the SQL Server

The `RAND()` function is a Non-Deterministic function but if we provide the seed value the function becomes deterministic as the same value gets returned for the same seed value.

Example1: `SELECT Rand(1)`

Deterministic as it returns the same value every time

Example2: `SELECT Rand()`

Non-deterministic as the returned values change every time

In the next article, I am going to discuss [Transaction Management in SQL Server](#) with Examples. Here, in this article, I try to explain the **Deterministic and Non-Deterministic Functions in SQL Server** with Examples. I hope this article will help you with your need. I would like to have your feedback. Please post your feedback, question, or comments about this article.